PHYSICAL CHEMISTRY CUMULATIVE EXAM

March 2, 1996

1. (a) An ESR spectrum for an organic radical hydrocarbon containing two carbon atoms consisted of 12 lines having intensities of 1,2,3,1,6,3,3,6,1,3,2,1. What is the radical?

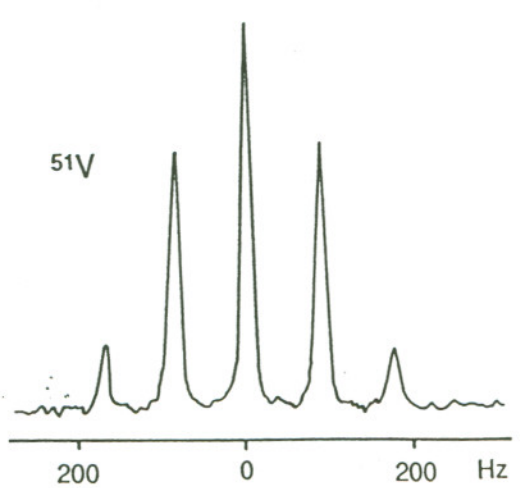
$I(C) = ½$, $I(H) = ½$, $I(e) = ½$

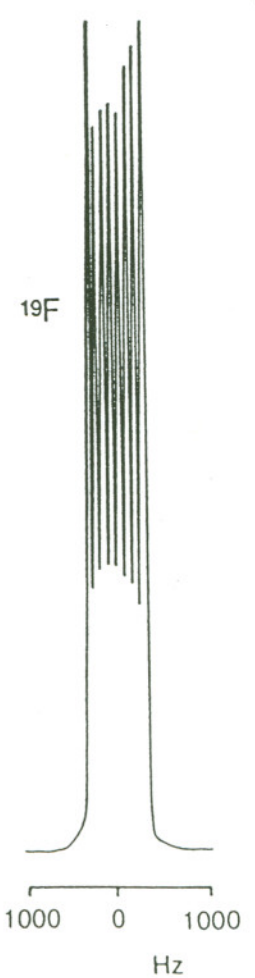
% $^{13}C = 1\%$ % $^{1}H = 99.9\%$

(b) The ^{51}V and ^{19}F spectra of the complex ion $(VOF_4)^-$ are shown below. What is the structure of the ion?

$I(^{51}V) = 7/2$ $I(^{19}F = ½)$

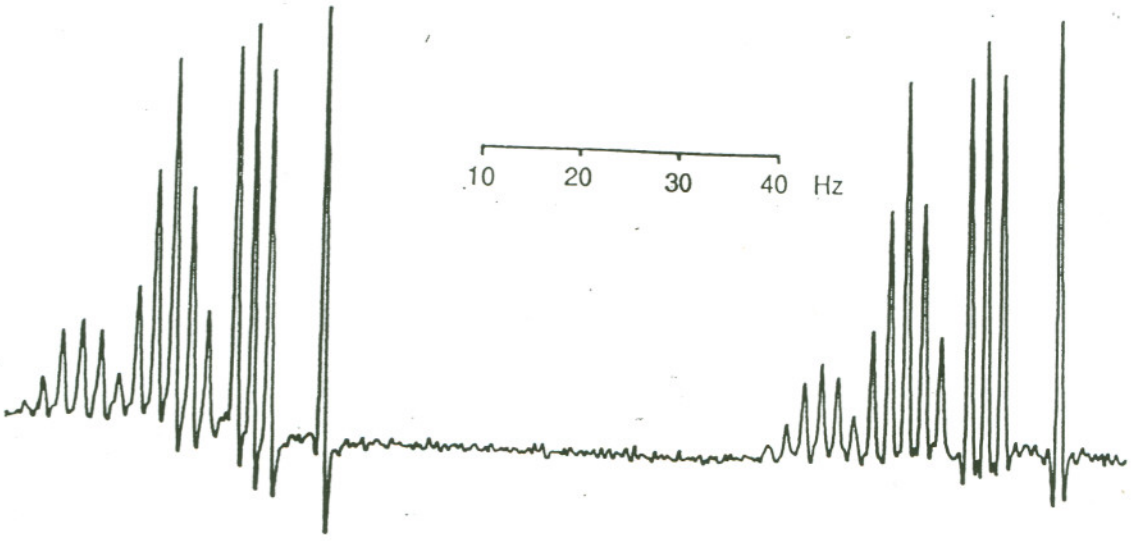
% $^{51}V = 99.8$ % $^{19}F = 100$





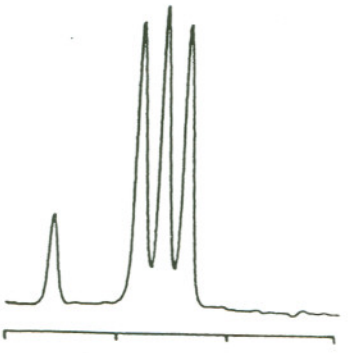
(c) ^{15}N-Labelled ammonium chloride (NH_4Cl) was dissolved in a mixture of H_2O and D_2O and then strongly acidified. Interpret as fully as possible the resulting ^1H spectrum shown below.

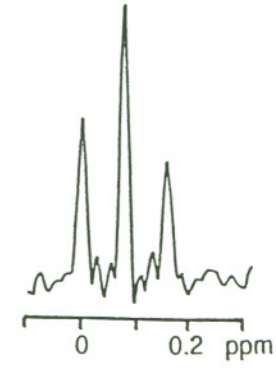
$I(^1H) = ½$ $I(^2H \text{ or } D) = 1$



(d) The 400 MHz (^1H) spectrum of a sample of labelled CD_2Cl_2 is shown below. Assign and explain the signals. CD_2Cl_2 usually consists of 99.8% deuterium.

$I(^2H \text{ or } D) = 1$ $I(^1H) = ½$



(e) The bacterium Nitromonas biosynthesizes 3-nitropropanoic acid, A, from NH_3. If the bacterium is grown on ^{15}N-labelled ammonia and natural-abundance oxygen gas ($^{16}O_2$), the ^{1}H-decoupled ^{15}N resonance of A is a sharp singlet; the shift of this singlet is arbitrarily assigned as zero. When the experiment is repeated with ^{15}N labelled ammonia and a 1:1 mixture of $^{18}O_2$ and $^{16}O_2$, A has the ^{15}N spectrum shown below. What conclusion can be drawn about the source and mechanism of oxygen incorporation?

A